

AI-DND

WORLD_SIMULATION_SYSTEM.md

Living World Simulation System

Version 0.1

1. DESIGN PHILOSOPHY

The world exists independently from the player.

The player does not activate the world.

The player enters an already living world.

The world continues to change:

- with the player;
 - without the player;
 - despite the player.
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2. WORLD SIMULATION LAYERS

The world consists of several simulation layers.

Individual Layer

Simulation of:

- NPCs;
 - creatures;
 - companions.
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Settlement Layer

Simulation of:

- villages;
 - towns;
 - cities;
 - camps.
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Regional Layer

Simulation of:

- economics;
 - wars;
 - diseases;
 - weather.
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Global Layer

Simulation of:

- factions;
 - trade routes;
 - migrations;
 - major events.
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3. TIME SYSTEM

Time exists continuously.

Units:

- minute;
- hour;
- day;
- week;
- month;
- year.

The world updates regardless of player actions.

4. DAILY SCHEDULES

Every NPC has:

- sleep time;
- work time;
- meal time;
- social time;
- personal time.

Example:

Blacksmith:

06:00 Wake up

07:00 Breakfast

08:00 Work

13:00 Lunch

14:00 Work

20:00 Tavern

22:00 Sleep

5. WEATHER SYSTEM

Weather affects:

- travel;
- combat;
- economy;
- health;
- agriculture.

Examples:

- rain;
- storms;
- snow;

- fog;
 - drought.
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6. SEASONS

Seasons change:

- environment;
 - prices;
 - diseases;
 - travel speed;
 - wildlife.
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7. ECONOMY

The economy is simulated.

Examples:

- production;
 - consumption;
 - shortages;
 - inflation;
 - trade.
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8. RESOURCES

Resources include:

- food;
- wood;
- iron;
- coal;
- herbs;
- black crystal.

Resource shortages affect society.

9. FOOD SYSTEM

Settlements require food.

Without food:

- prices rise;
 - starvation begins;
 - migration occurs;
 - riots happen.
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10. TRADE ROUTES

Trade routes transport:

- food;
- weapons;
- medicine;
- luxury goods.

Routes may be:

- protected;
 - attacked;
 - destroyed.
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11. CARAVANS

Caravans:

- travel;
- trade;
- disappear;
- get robbed.

The player may:

- escort;
 - rob;
 - protect;
 - ignore.
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12. WAR

Wars occur dynamically.

Effects:

- deaths;
 - shortages;
 - refugees;
 - economic collapse.
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13. FACTIONS

Factions possess:

- goals;
- enemies;
- allies;
- resources;
- territory.

Factions act independently.

14. SETTLEMENT DEVELOPMENT

Settlements evolve.

Examples:

Village:

Population: 100

After prosperity:

Population: 300

New buildings:

- market
- walls

- barracks
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15. SETTLEMENT DECLINE

Settlements may collapse.

Reasons:

- war;
 - famine;
 - disease;
 - monsters;
 - economics.
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16. DISEASE SPREAD

Diseases spread through:

- people;
- trade;
- animals;
- water.

Effects:

- death;
 - panic;
 - migration.
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17. MIGRATION

NPCs relocate.

Reasons:

- war;
 - famine;
 - opportunity;
 - fear.
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18. CRIME

Crime changes dynamically.

Examples:

- theft;
- smuggling;
- murder;
- corruption.

Crime influences society.

19. WILDLIFE

Animals:

- migrate;
- reproduce;
- hunt;
- die.

Examples:

- wolves;
 - deer;
 - bears.
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20. MONSTERS

Monster populations change.

Examples:

Too many wolves:

- fewer deer;
- more attacks.

Dragon appears:

- migration;

- fear;
 - economic collapse.
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21. AGING

NPCs age.

Effects:

- appearance;
 - health;
 - memory;
 - status.
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22. BIRTH

Children are born.

Effects:

- population growth;
 - inheritance;
 - social change.
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23. DEATH

NPCs die.

Reasons:

- age;
- disease;
- combat;
- accidents;
- murder.

Death changes the world.

24. SUCCESSION

When leaders die:

- heirs inherit;
 - civil wars begin;
 - factions change.
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25. CULTURE

Settlements develop culture.

Examples:

- traditions;
 - religion;
 - cuisine;
 - clothing.
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26. RUMOR NETWORK

Information spreads.

Speed depends on:

- roads;
 - merchants;
 - politics;
 - witnesses.
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27. PLAYER IMPACT

The player may:

- save settlements;
- destroy settlements;
- change wars;
- change economies;
- alter history.

28. HISTORY

The world remembers.

Examples:

- wars;
- disasters;
- heroes;
- crimes.

Books may be written.

Songs may be composed.

29. WORLD EVENTS

Examples:

- plague;
 - rebellion;
 - invasion;
 - famine;
 - discovery;
 - prophecy.
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30. SIMULATION DISTANCE

Not all regions simulate equally.

Close to player: high simulation.

Far from player: abstract simulation.

This preserves performance.

31. AI NPC INTEGRATION

NPCs react to:

- economy;
 - weather;
 - rumors;
 - war;
 - disease;
 - politics.
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32. PLAYER ABSENCE

If the player ignores problems:

the world still changes.

Example:

Village requests help.

Player refuses.

One month later:

Village no longer exists.

33. SAVE SYSTEM

The world state persists.

Nothing resets.

34. CORE RULE

The player lives inside the world.

The world does not live for the player.